

CHEATSHEET · QUICK REFERENCE

Lab 32 — Network Connectivity Troubleshooting

Work the OSI stack bottom-up: link, ARP, routing, reachability, DNS and port tests, capture packets for the silent direction and read VLAN/NIC indicators.

Layer 1-3

```
ip -br link
```

Link state (Link detected)

```
ethtool eth0 | head
```

Speed, duplex, link

```
ip neigh
```

ARP table (gateway FAILED = L2 issue)

```
ip route get 8.8.8.8
```

Which route/gateway is chosen

Reachability and DNS

```
ping -c 3 8.8.8.8
```

Reach a public IP

```
traceroute -n 8.8.8.8
```

Path to the destination

```
mtr -rwc 5 8.8.8.8
```

Loss + latency per hop

```
dig +short example.com
```

Resolve a name

Ports and capture

```
nc -vz example.com 443
```

Test a single port

```
nmap -p 22,80,443 -Pn 127.0.0.1
```

Scan a range

```
tcpdump -ni eth0 -c 10 'icmp or port 53'
```

Capture the silent direction

```
ss -tulnp
```

Listening sockets

Cause indicators

Cause	Indicator
Misconfigured VLAN	Has IP but can't reach gateway
Port security	Link up then drops in seconds
Speed/duplex mismatch	High error counters (ip -s link)
Wrong subnet mask	Local OK, distant hosts fail
DHCP failure	APIPA 169.254.x.x address